MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM 23 VOLTS, 6 AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery assembly described herein shall consist of this document and the latest issue of MIL-B-81757.

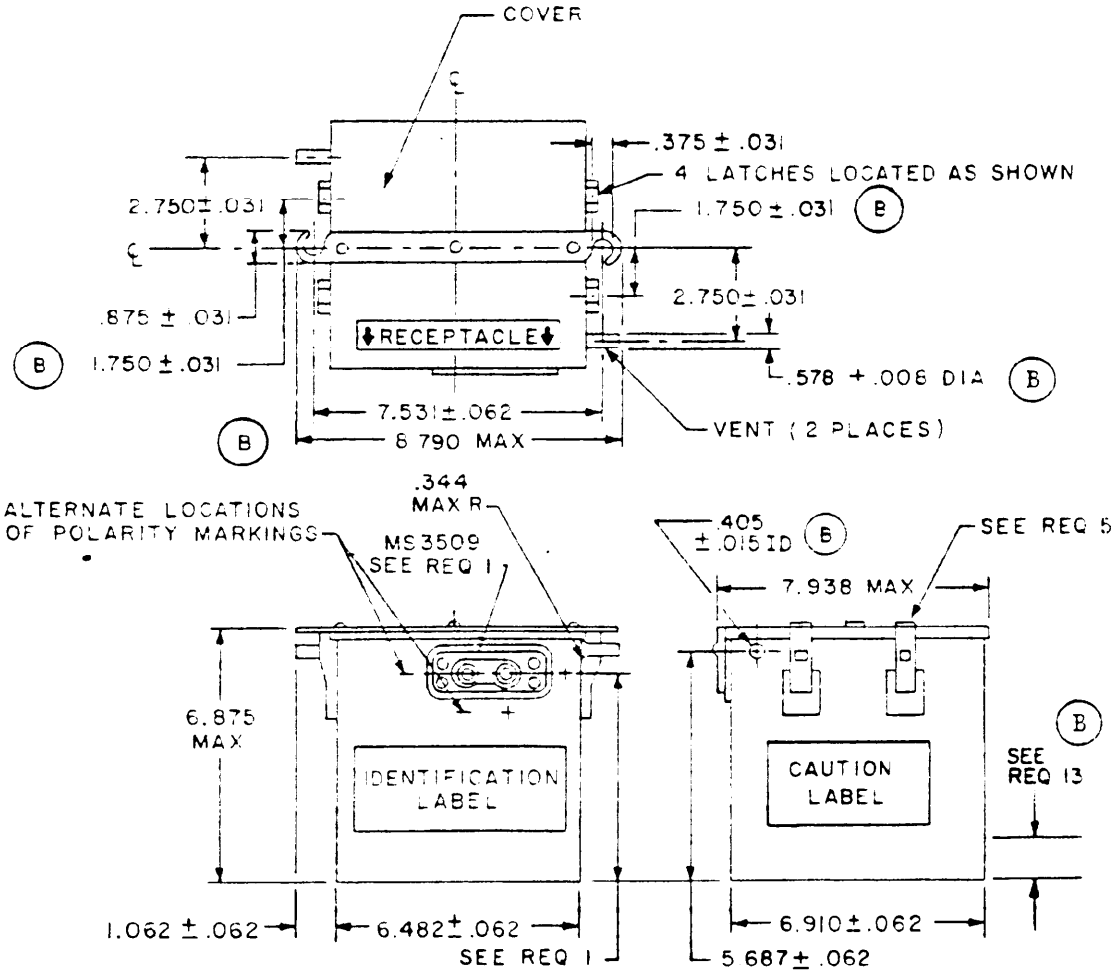


Figure 1. Battery assembly P/N M81757/10-1.

FSC 6140

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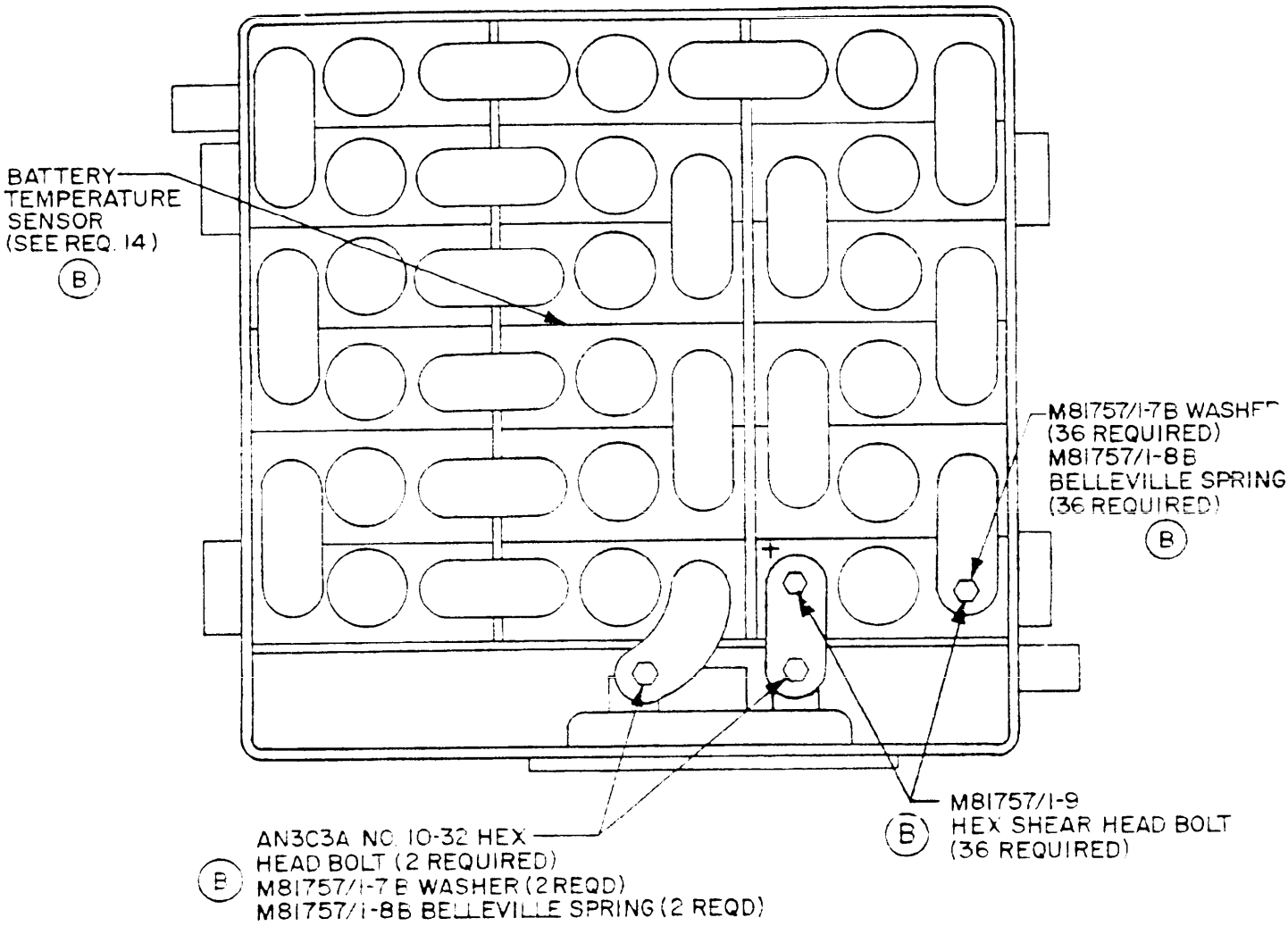
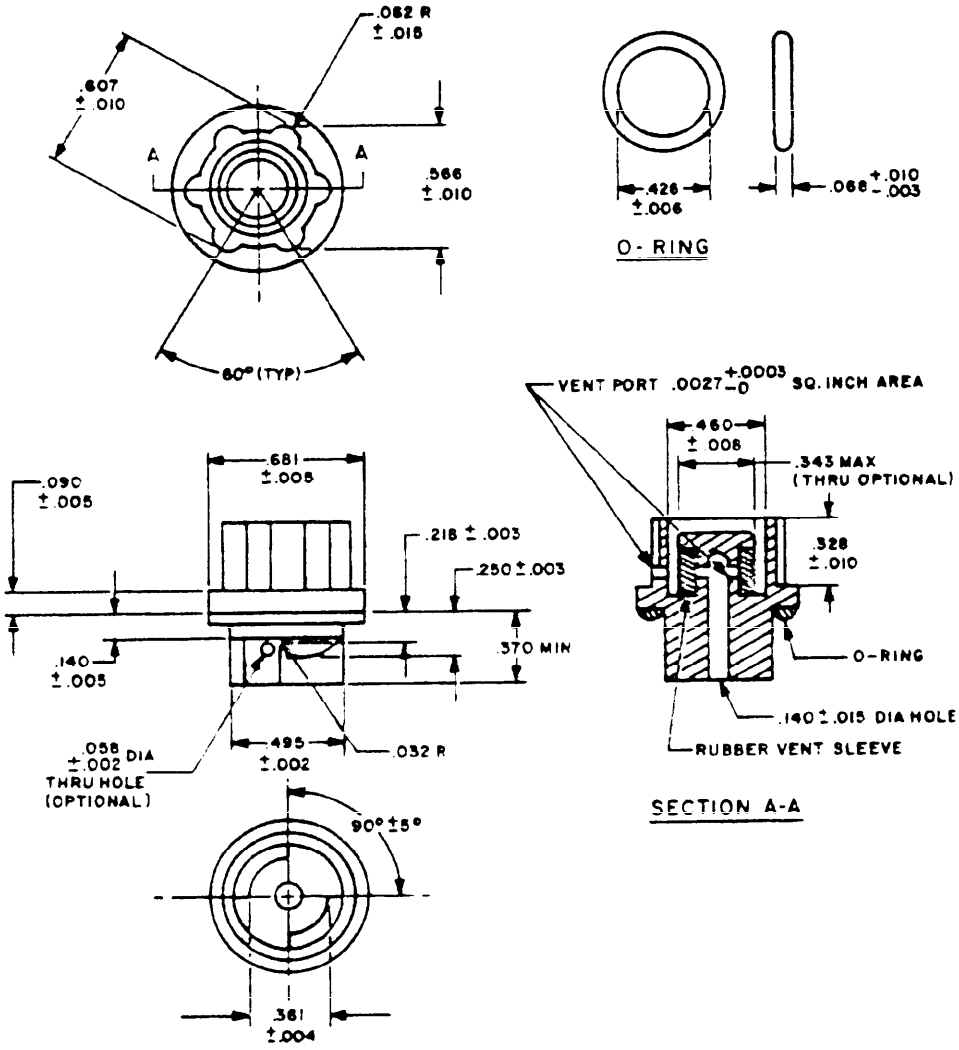


Figure 2. Cell arrangement.

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(B) Figure 3. Vented filler cap assembly.

REQUIREMENTS:

1. Filler cap shall be polyamide per ASTM STD D4066-82, Composition A, Type III.
(B) 2. Cap shall be of molded one-piece construction.
3. O-ring shall be of ethylene propylene rubber.

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REQUIREMENTS:

1. The receptacle terminal block shall be 0.866 ± 0.015 from the centerline of the contact pins to the top surface of the terminal block. The receptacle will be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle shall conform to MS3509-28.

(B) 2. All dimensions are prior to epoxy coating except: overall length - 8.790 max., overall width - 7.938 max., and overall height - 6.875 max.

(B) 3. The cell filler cap shall be in accordance with Figure 3.

(B) 4. Spacers if used shall be polyethylene sheet per ASTM STD D4066-82, Comp A, Type I and Type III, minimum 0.015 inch thick, color natural.

5. Latches and strikes shall not be epoxy coated.

6. The battery voltage shall be 23 volts (nominal).

7. The battery weight shall be 24 pounds maximum when filled.

8. The battery shall contain 18 cells arranged as shown in Figure 2.

9. The rated capacity at 1C rate shall be 6 A.H./1 HR/80°F/17.0V.

10. All cells within a battery shall be from the same production lot.

(B) 11. Cell removal shall be impeded by using the M81757/1-9 bolt to secure the cell connector straps.

12. Battery part number M81757/10-1 shall conform to MIL-B-81757 except as modified by the following:

3.5.7 Cells. Delete.

3.5.7.1 Mountings. Delete

3.6.5.2.1 Cell terminal marking. Delete last sentence.

3.6.5.2.2 Cell identification. Delete.

(B) 3.6.11 Temperature rise and float. Change second sentence to read: The charging currents shall not exhibit increasing values of more than a total of 2 amperes for a 6 ampere-hour battery, and a 10 ampere-hour battery, 4 amperes for a 20 ampere-hour battery, and 7 amperes for a 30 ampere-hour battery at any time during the charge.

3.6.12 Cycling test.

a. Change "14.0 volts" to "13.3 volts."
b. Change "12.0 volts" to "11.4 volts."

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TABLE II. Performance requirements.

3/ Change "17.0 volts" to "16.1 volts."
4/ Change "13.0 volts" to "12.3 volts."
5/ Change "12.0 volts" to "11.4 volts."

3.6.5.2.3 Battery identification. Each battery shall have its identification marking silk screened with an epoxy overlay, or equal, that is impervious to 1.3 ± 0.025 specific gravity KOH electrolyte, in accordance with applicable specifications sheet. Marking shall be in white print per MIL-M-13231. The marking shall show the following:

INFORMATION REQUIRED	EXAMPLE
Battery, Storage	Battery, Storage
Military Battery Part No.	M81757/10-1
Replaces (include replacement data from applicable battery specification sheet)	Replaces MS90447-2
National Stock Number	NSN 6140-01-***-****
Voltage (Nominal)	23 Volts
Maximum Weight	Max. Wt. 24 Lbs.
Military "C" Rated Capacity	6 A.H./1 HR/80°F/17.0V
Contract Number	***
Date Code (see 3.6.5.2.5)	Date Code ***
(B) Manufacturer's Name	***
Manufacturer's Part Number	***
(B) Manufacturer's Serial Number	***

3.6.5.2.4 Battery caution and receptacle markings. These markings, in accordance with the applicable specification sheet, shall be in white printing in accordance with MIL-M-13231. They shall be impervious to 1.3 ± 0.025 specific gravity KOH electrolyte. They shall be located as shown on the applicable specification sheet and read as follows:

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↓ RECEPTACLE ↓	0.50 ± 0.03 Inch Lettering Height
"CAUTION"	0.50 ± 0.03 Inch Lettering Height
THE CELLS IN THIS BATTERY ARE NOT TO BE REMOVED.	0.375 ± 0.03 Inch Lettering Height
THIS BATTERY SHALL BE SERVICED ONLY IN AN AUTHORIZED NICKEL-CADMIUM BATTERY SHOP.	0.375 ± 0.03 Inch Lettering Height

(B)

5.5 Marking. The individual battery package shall be marked on one of the two largest vertical surfaces with the supersession data from note 2.

The lettering shall be at least 1/2 inch high on the battery package. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

(E) 13. The base dimensions of the container shall be within tolerance for a distance of 1.4 inches up from the bottom.

(E) 14. A sensor (thermocouple or thermistor), when required shall be located as shown in Figure 2 for the purpose of monitoring battery temperature during testing to this specification. The sensor shall be positioned in the approximate center of the cell container's widest outer surface (equidistant from the top and bottom and from side to side).

NOTES:

1. All dimensions are in inches, unless otherwise specified.

(B) 2. This specification sheet supersedes MS90447-2 and MS90321 part numbers as follows, when a product has been qualified for M81757/10-1 part number.

Part Number	Replaces Part Number
M81757/10-1	MS90447-2, MS90321-84W

Part Number	Superseded By
MS90447-2, MS90321-84W	M81757/10-1

Preparing activity:
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